

Assignment $\rightarrow \mu_0 \mu_r = \mu = (\phi) \times \text{length} \therefore$

1) A circuit iron ring of mean diameter 25 cm and cross sectional area 9 cm^2 is wound with a coil of 100 turns and carries a current of 1.5 A . The relative permeability of iron is 2000. Calculate the amount of flux produced in the ring.

mean diameter, $d = 25 \text{ cm} = 0.25 \text{ m}$

cross sectional Area, $A = 9 \text{ cm}^2 = 9 \times 10^{-4} \text{ m}^2$

Current, $I = 1.5 \text{ A}$

Relative permeability, $\mu_r = 2000$

μ_0 (permeability of space) $= 4\pi \times 10^{-7} \text{ H/m}$

Now, mean mag. path length, $l = \pi d$

$$l = \pi d = 0.785 \pi \text{ m}$$

Magnetomotive force $= NI = 150 \text{ A-turns}$

magnetic field intensity, $H = \frac{NI}{l} = \frac{150 \text{ A/m}}{0.785 \pi}$

magnetic flux density, $B = \mu_0 \mu_r H = \mu_r \frac{NI}{l}$

$$\therefore \text{Flux } (\phi) = BA = \mu_0 \mu_r \frac{NI^2 A}{l}$$

$$\Rightarrow \phi = \frac{(4\pi \times 10^{-7}) \times 2000 \times 1000 \times 1.5 \times 9 \times 10^{-4}}{0.25\pi}$$

$$\Rightarrow \phi = 4.32 \times 10^{-4} \text{ Wb (Ans)}$$

2) A 40 KVA 3200/400 V, single phase, 50 Hz transformer has 112 turns on the secondary winding. Calculate the number of turns on the primary winding. What is the secondary current at full load? What should be cross-sectional area of the core flux density of 1.2 Wb/m²?

Transformer rating, $S = 40 \text{ KVA}$

Primary voltage, $V_p = 3200 \text{ V}$

secondary voltage, $V_s = 400 \text{ V}$

freq, $f = 50 \text{ Hz}$

secondary turns, $N_s = 112$

desired core flux density, $B = 1.2 \text{ Wb/m}^2$

(a) turns on primary coil, $N_p \rightarrow = \phi$

using turns ratio, $\frac{N_p}{N_s} = \frac{V_p}{V_s}$

$$\Rightarrow N_p = N_s \times \frac{V_p}{V_s} = 112 \times \frac{3200}{400}$$

$$\therefore N_p = 896 \text{ turns (Ans)}$$

(b) Secondary current at full load, $I_s \rightarrow$

$$I_s = \frac{S}{V_s} = \frac{40000}{400} = 100 \text{ A (Ans)}$$

(c) core cross-sectional area for given flux density $\rightarrow$

Flux in core per cycle is related to induced RMS voltage by,

$$E = 4.44 f N \phi \text{ (for sinusoidal flux)}$$

$$\text{Now, } \phi = \frac{V_s}{4.44 f N_s} \text{ (using secondary values)}$$

$$\phi = \frac{400 \times 1.44 \times 50 \times 1.2}{4.44 \times 50 \times 1.2} \quad (P)$$

$$4.44 \times 50 \times 1.2$$

$$\Rightarrow \phi = 0.0160875 \text{ Wb}$$

$$\text{Now, } \lambda_{\text{core}} = \frac{\phi}{B} = \frac{0.016085}{1.2}$$

$$(A) \text{ } \lambda_{\text{core}} = 0.01341 \text{ m}^2 = 134.1 \text{ cm}^2$$

← 20 kVA 11kV to 220V transformer (Ans) (d)

eg) The frequency of a rotor induced EMF of 400V, three phase, 50 Hz, six pole induction motor is 2 Hz. Calculate the speed of the motor.

supply frequency, $f_s = 50 \text{ Hz}$ (c)

poles, $P = 6$

Rotor-induced frequency = 2 Hz

(x) $f_r = f_s - \frac{P}{120} \times N$

Now, synchronous speed, $N_s = \frac{120 f_s}{P}$
(in rpm)

$$N_s = \frac{120 \times 50}{6}$$

$$\Rightarrow n_s = 1000 \text{ rpm}$$

$$\text{slip, } s = \frac{f_r}{f_s} = \frac{2}{50} = 0.04 = 4\%$$

$$\text{Now, Rotor speed, } n = (1-s)n_s$$

(speed of motor)

$$\Rightarrow n = 0.96 \times 1000 = 960 \text{ rpm (Ans)}$$